

simulations generate millions of different scenarios that estimate the probabilities of earning various return levels, both with and without leverage. The tables that follow report the probabilities for a variety of outcomes.

TABLE 4.2 Leverage: The Good—What Are the Chances of Doubling the Market? (Market Expected Return: 15%; Volatility: 30%)

Leverage Ratio	Interest Cost		
	3%	5%	7%
0%	30.9%	30.9%	30.9%
25%	37.4%	36.9%	36.4%
50%	42.1%	41.2%	40.3%
100%	48.0%	46.7%	45.4%
250%	55.7%	53.8%	51.9%

Source: Moshe Milevsky and the IFID Centre, 2008.

Here’s how to read Table 4.2. Assume that you are paying 5% interest on your margin account. If you leverage at a rate of 100%—one borrowed dollar for every dollar of original equity investment capital—then the odds of earning double the market rate (that is, $2 \times 15\% = 30\%$) on your investment are 46.7%. In other words, the chances of converting \$10,000 into \$13,000 are 46.7%. That’s slightly less than a one-in-two chance of doubling the market.

Compare that to the chances of earning the same 30% ($2 \times 15\%$) if you don’t leverage at all. Then, as Table 4.2 illustrates, your zero-leverage chances are a mere 30.9%. This is less than one chance in three. Notice the huge increase in the probability of doing very well if you leverage. Sound tempting? Many have been lured.

Of course, as you can see from Table 4.2, if you have to pay more than 5% on your margin account, the odds of doing better than doubling the market will go down a bit. That’s simply because the market has to earn even more, which is even less likely, to pay back your higher-interest borrowing. Likewise, if your interest cost is lower than 5%, your odds are somewhat better, but not by much. The real improvement in odds comes from higher amounts of leverage.